

Influence of Quiet Rest and Acute Aerobic Exercise Performed in a Naturalistic Environment on Selected Psychological Responses

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The primary purpose of the present study was to examine psychological responses of 50 young women following single episodes of aerobic exercise (AE) performed in a naturalistic exercise setting and quiet rest (QR). Given the salience of self-presentational qualities in a naturalistic exercise setting, a secondary purpose was to examine whether psychological responses to acute exercise varied as a function of social physique anxiety (SPA). The dependent measures of state anxiety, positive engagement, revitalization, tranquility, and physical exhaustion were assessed immediately prior to and then at 5 and 30 minutes following each condition. It was found that (a) AE and QR were associated with decreases in state anxiety and increases in tranquility; (b) AE was associated with increases in positive engagement and revitalization; and (c) these changes did not vary as a function of SPA level.

Key words: feeling states, social physique anxiety, state anxiety

Mounting evidence indicates that exercise is associated with improvements in an array of mental health outcomes. Specifically, acute bouts of aerobic exercise have been linked with increases in positive affect (McAuley, Talbot, & Martinez, 1999; Tate & Petruzzello, 1995), revitalization (Bozoian, Rejeski, & McAuley, 1994; Gauvin & Rejeski, 1993), and energetic arousal (Thayer, 1987), and to reductions in state anxiety (Breus & O'Connor, 1998; Raglin & Morgan, 1987) and negative affect (Petruzzello, Jones, & Tate, 1997). The recent proliferation of exercise-affect research provides considerable evidence to support the psychological benefits of acute exercise. However, a comprehensive understanding of the factors that contribute to these improvements remains elusive.

One cognitively based explanation for the anxiolytic effect of exercise, the distraction or time-out hypothesis, emanated from the observation of reductions in state anxiety following acute bouts of aerobic exercise, meditation, and quiet rest (Bahrke & Morgan, 1978; Raglin & Morgan, 1987). The primary tenet of the time-out hypothesis is that it is the distraction from one's daily stressors provided by these activities that is responsible for the observed anxiolysis. Empirical support

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for the distraction hypothesis has been equivocal, however. For example, while comparable anxiolytic responses have been documented following exercise and quiet rest (Bahrke & Morgan, 1978; Raglin & Morgan, 1987; Youngstedt, O'Connor, Crabbe, & Dishman, 1998), reductions in state anxiety following quiet rest have not emerged in other investigations (Boutcher & Landers, 1988; McAuley, Mihalko, & Bane, 1996; Tate & Petruzzello, 1995). Although some researchers have attributed the disparity in findings to behavioral artifacts (Morgan, 1997; Raglin, 1997), others have interpreted the inconsistent findings as evidence refuting the efficacy of the distraction hypothesis (McAuley et al., 1996). Consequently, the prevailing ambiguity has precluded the development of a consensus regarding the anxiolytic benefits of acute exercise and quiet rest.

The social environmental setting of the exercise stimulus has also been advanced as an explanation for the psychological benefits of an acute bout of exercise (Gauvin & Brawley, 1993; Gauvin & Rejeski, 1993). Specifically, Rejeski and his colleagues contend that improvements in psychological states following acute exercise are related to the unique social context of exercise settings (Gauvin, Rejeski, & Norris, 1996; Rejeski, 1994; Rejeski, Gauvin, Hobson, & Norris, 1995). For example, in a comparison of naturalistic and laboratory exercise conditions, Gauvin and Rejeski (1993) observed increases in positive engagement and revitalization following exercise in the naturalistic setting only. Consistent with these findings, improvements in positive feeling states have been shown to be sensitive to both the social environmental context during acute exercise (Gauvin et al., 1996; Rejeski et al., 1995) and to the leadership style employed in group exercise settings (Turner, Rejeski, & Brawley, 1997). In short, these findings illustrate the salience of the exercise environment in generating positive feeling states.

Conversely, reductions in state anxiety have been reported regardless of the exercise setting (Breus & O'Connor, 1998; Raglin & Morgan, 1987; Youngstedt et al., 1998). In an examination of the influence of exercise environment on state anxiety, McAuley et al. (1996) observed anxiolytic responses following acute exercise performed in both naturalistic and laboratory settings. However, they also found that state anxiety was higher at each assessment interval in the laboratory condition. McAuley et al. suggested the novelty of laboratory testing environments may provoke anxiety in some individuals, perhaps resulting in an exaggerated anxiolytic effect. By contrast, Petruzzello (1995) demonstrated that state anxiety reported in the laboratory setting prior to exercise was lower than anxiety reported during the course of normal daily activities in a sample of male college students. Hence, in light of the discordant findings, additional inquiry is needed to elucidate the role of the environment in postexercise anxiety and feeling-state responses.

Although naturalistic exercise settings have been associated with the enhancement of positive feeling states, they are also replete with qualities that give rise to self-presentational concerns. Thus, the same social environmental aspects that help generate positive feeling states can also be interpreted as threatening by those individuals concerned about negative social evaluation (Conroy, Motl, & Hall, 2000; Leary, 1992). Bain and colleagues (Bain, Wilson, & Chaikind, 1989) found that obese women reported anxiety at the prospect of negative social evaluation as the greatest deterrent to exercising in public. Consonant with such concerns, social physique anxiety (SPA; Hart, Leary, & Rejeski, 1989), the anxiety felt when one believes others are negatively evaluating his or her physique, has been found to be related to environmental aspects of the exercise setting. For

example, women with high SPA reportedly prefer exercising in private settings (Spink, 1992) and in exercise environments in which less revealing attire is worn (Crawford & Eklund, 1994). Furthermore, Katula and colleagues (Katula, McAuley, Mihalko, & Bane, 1998) observed that, when compared to a non-mirrored setting, women with high SPA reported lower self-efficacy during a bout of cycling performed in front of a full length mirror. Using similar methodology, Willow and Mihalko (2000) demonstrated that SPA also predicted affective responses during a bout of acute resistance exercise performed in front of a full-length mirror.

The emerging consensus, that self-presentational factors influence exercise behaviors (Leary, 1992), led Gauvin et al. (1996) to suggest that systematic covariation of characteristics related to the social context of the exercise environment would contribute to a greater understanding of the exercise/affect relationship. The findings of Katula et al. (1998) and Willow and Mihalko (2000) provide evidence that SPA interacts with the evaluative nature of the exercise environment to influence self-efficacy and affective states during acute physical activity. Thus, further exploration of the impact of SPA on exercise-induced feeling states may promote a greater understanding of the factors that influence the relationship between exercise and psychological well-being.

To summarize, due to both the equivocal findings of anxiolysis following quiet rest and the limited knowledge of the influence of quiet rest upon positive feeling states, more research is warranted that compares the effect of acute exercise and quiet rest on anxiety and positive affect responses. Thus the primary purpose of the present investigation was to contrast the efficacy of acute bouts of aerobic exercise performed in a naturalistic environment with quiet rest on the amelioration of subjective psychological states (i.e., state anxiety and feeling states). Given the salience of self-presentational aspects in a naturalistic exercise setting (Conroy et al., 2000; Leary, 1992), a secondary purpose was to examine whether psychological responses to exercise are moderated by initial SPA level. It was hypothesized that both aerobic exercise and quiet rest would be associated with improvements in state anxiety and feeling states. However, due to the prevalence of self-presentational qualities in many naturalistic exercise settings (mirrored walls, presence of other exercisers, etc.), it was also hypothesized that for participants with high SPA, there would be no improvements in psychological states following exercise.

Method

Participants and Measures

Participants were 50 female university students, mean age 19.9 years ($SD = 1.6$ yr); 92% were Caucasian. An exclusively female sample was employed based on findings that the intensity and direction of SPA differs as a function of gender (Hart et al., 1989).

Social Physique Anxiety Scale (SPAS). The SPAS was developed by Hart et al. (1989) as a trait measure of self-presentational anxiety related to physique. Although it was originally constructed as a 12-item measure, researchers have advocated the use of a 9-item version that is more representative of the SPA construct (Martin, Rejeski, Leary, McAuley, & Bain, 1997). The 9-item SPAS was used in the present investigation; it had satisfactory internal consistency

($\alpha = .86$). Participants responded to the questionnaire by indicating the degree to which each item was characteristic or true of them on a Likert scale ranging from 1 (*not at all*) to 5 (*extremely*). Items 5 and 8 were reverse scored and then all the item responses were summed to create an overall SPAS score that could range from 9 to 45.

State-Trait Anxiety Inventory (STAI). State anxiety was measured using Form Y-1 (SAI) of the 20-item STAI (Spielberger, Gorsuch, Lushene, Vagg, & Jacobs, 1983). The SAI has undergone rigorous validation procedures (Spielberger et al., 1983) and has demonstrated adequate internal consistency ($\alpha = .89$) within the present sample. Participants responded to each item on a 4-point Likert scale ranging from 1 (*not at all*) to 4 (*very much so*).

Exercise-Induced Feeling Inventory (EFI). The EFI is a 12-item multidimensional affective measure that requires participants to indicate on a 5-point scale ranging from 0 (*do not feel*) to 4 (*feel very strongly*) the extent to which they are experiencing four distinct feeling states: positive engagement, revitalization, tranquility, and physical exhaustion (Gauvin & Rejeski, 1993). The EFI has been shown to have adequate psychometric properties (Gauvin & Rejeski, 1993; Gauvin et al., 1996) and it demonstrated satisfactory internal consistency scores within the present sample (α values $>.86$ for each subscale). Means for each subscale were obtained by summing the scores for the items that comprise each subscale.

Leisure-Time Exercise Questionnaire (LTEQ). The LTEQ measures the self-reported frequency of strenuous, moderate, and mild bouts of exercise during the course of a typical week (Godin & Shephard, 1985). LTEQ subscale scores are converted into metabolic equivalents (METs; $[\text{Mild} \times 3] + [\text{Moderate} \times 5] + [\text{Strenuous} \times 9]$) and summed to provide an estimate of total METs expenditure from exercise for an average week. Previous research has shown that the LTEQ possesses adequate reliability and validity (Jacobs, Ainsworth, Hartman, & Leon, 1993).

Body Composition. Two measures of body composition were obtained. First, percent body fat was assessed with skinfold calipers using the three-site technique (triceps, suprailiac, quadriceps; Jackson, Pollock, & Ward, 1980). The average of the three skinfold measurements obtained from each site was converted to an estimate of body composition using the generalized equations developed by Jackson et al. (1980). Second, body mass index (BMI) was calculated from direct height and weight assessments.

Ratings of Perceived Exertion (RPE). In order to quantify the participants' overall effort sense during the exercise session, RPE were obtained using Borg's 6–20 scale (Borg, 1973). RPE has been conceptualized as a psychobiological configuration of sensations associated with the perception of effort during exercise and has been used extensively in previous research (Borg, 1998).

Procedures

Prior to participation, each volunteer read and signed an informed consent document approved by the Institutional Review Board. Upon completing the study requirements, all volunteers received extra class credit for their involvement. In order to examine the influence of SPA on psychological responses, all volunteers completed the SPAS prior to participation. Only those scoring 36 or above and 22 or below were eligible to be included in the study. The SPAS values for eligibility

were determined based on average SPAS scores of women reported in previous research (Hausenblas & Martin, 2000; Martin et al., 1997). Thus the sample was composed of women with high ($n = 25$) or low ($n = 25$) SPA. Each participant completed one 20-min episode of aerobic exercise and quiet rest. Episodes were completed on separate days and the order of each condition was randomly assigned and counterbalanced. Assessments of state anxiety and feeling states were obtained immediately prior to, immediately following, and again 30 minutes following each condition.

In order to standardize the clothing worn during each condition to avoid the use of clothing as part of protective (loose, baggy attire) or revealing (tight or scant attire) self-presentational strategies, participants were asked to wear a T-shirt and shorts at each session. Prior to the initial session, participants also completed the LTEQ and a medical history questionnaire in order to screen for potential contraindications for exercise. Upon arriving at the laboratory, they completed the baseline assessments of the STAI and EFI. In order to minimize the potential influence of demand characteristics or expectancy effects, participants were not informed as to the purpose of the study, and following the completion of the baseline measures they were only told which condition they would be performing during a particular session.

Exercise Condition. Exercise sessions were conducted at a university fitness facility next to the Exercise Psychology Laboratory. The 4,700-sq-ft fitness center is furnished with aerobic and resistance exercise equipment and is a popular recreational exercise facility used by both male and female students. While the testing was being conducted, an average of 25 to 30 exercisers and 3 staff members were present in the facility. Thus the fitness center was a naturalistic exercise environment that provided ample opportunity for both the realization of self-presentational concerns and social interaction via the constant presence of other exercisers, staff members, and mirrored walls. Participants were instructed to exercise for 20 minutes at their preferred level of exertion on the aerobic exercise equipment of their choice (i.e., cycling, rowing, or stairmaster machines). They were permitted to utilize their preferred mode and level of exertion in order to provide a more ecologically valid assessment of the influence of acute exercise on psychological states.

A laboratory assistant escorted participants to the exercise facility. However, to prevent the researcher from serving as a buffer to either the social context or evaluative nature of the exercise environment, the assistant did not remain in the fitness center during the 20-min exercise session. At the conclusion of the session, participants recorded their RPE and returned to the exercise psychology laboratory. They then sat quietly in a room free of distractions for the next half hour, during which time the postexercise assessments were obtained.

Quiet Rest Condition. During the quiet rest session the participants sat quietly for 20 minutes in a room free of distractions in the exercise psychology laboratory. They were not given any specific instructions beyond being asked not to sleep or work during the session. Following the quiet rest session, body composition assessments were obtained. The potential influence of this evaluation on the dependent measures was addressed by instructing participants at the outset of the study that they could receive a body composition assessment if they so desired. Each participant was given this option after the quiet rest condition and all agreed to participate in the evaluation.

Analysis

Using a moderate effect size, the sample size employed provided a power that exceeded 0.80 for the analyses conducted (Tran, 1997). State anxiety and EFI data were analyzed with a 2 (Condition; exercise and quiet rest) $\times$ 3 (Time; pre, post, post-30 assessments) multivariate analysis of covariance (MANCOVA). Consistent with the recommendations of Gauvin et al. (1996), the SPAS score was entered into the MANCOVA as a covariate. Univariate analysis of variance (ANOVA) with Bonferroni adjustments ($p < .01$) were employed for follow-up analyses, and Tukey's post hoc tests were used to locate mean differences when significant effects were detected. Effect sizes (ES) were calculated by subtracting the post condition means from the baseline values and dividing by the pooled standard deviation. Although not central to the primary purpose of the investigation, differences among the high and low SPA participants in mild, moderate, strenuous, and total LTEQ scores were analyzed with a one-way ANOVA, and independent t -tests with alpha adjustments ($p < .01$) were used to analyze the % body fat, BMI, and RPE data.

Results

Consistent with previous research (Hart et al., 1989), high SPA participants were found to be characterized by significantly higher % body fat, $t(48) = 4.65$, $p < .001$, and BMI, $t(48) = 4.91$, $p < .001$, compared to low SPA participants. No significant differences between the high and low SPA groups were observed for mild, moderate, strenuous, or total LTEQ scores ($p > .06$). Additionally, no significant differences in RPE, $t(48) = 2.0$, $p > .05$, emerged between the high and low SPA groups during the exercise session. Therefore, the high and low SPA groups did not differ on exercise involvement during a typical week or the perception of effort during the exercise condition (see Table 1 for descriptive statistics).

Table 1 Body Composition, Ratings of Perceived Exertion, and Leisure Time Physical Activity (LTEQ) for High and Low SPA Participants ($M \pm SD$)

Variable	High SPA ($n = 25$)	Low SPA ($n = 25$)
Social Physique Anxiety	39.0 (2.7)*	18.0 (3.0)
% Body Fat	22.8 (2.5)*	19.2 (2.8)
Body Mass Index	24.1 (2.9)*	20.9 (1.5)
Ratings of Perceived Exertion	13.2 (1.5)	14.0 (1.5)
Leisure-Time Exerc. Questionnaire		
Mild	6.8 (7.3)	3.4 (5.1)
Moderate	8.2 (9.1)	8.8 (8.3)
Strenuous	19.8 (10.1)	21.9 (12.7)
Total	34.8 (19.1)	34.1 (16.9)

*Significantly higher than low SPA participants, $p < .001$

MANCOVA analyses revealed significant main effects for Time (Wilks' lambda = .87), $F(5, 88) = 8.23, p < .001$, and Condition (Wilks' lambda = .68), $F(5, 88) = 5.72, p < .001$, as well as a significant Condition $\times$ Time interaction (Wilks' lambda = .94), $F(10, 88) = 8.68, p < .001$. In order to evaluate whether the psychological responses were influenced by SPA, the interaction effects involving SPA were also evaluated. Neither the SPA $\times$ Condition ($p = .85$) nor the SPA $\times$ Time ($p = .26$) interactions were significant. Thus, psychological responses did not vary as a function of SPA. Follow-up univariate analyses revealed that state anxiety and the positive engagement, revitalization, and tranquility subscales of the EFI were responsible for the significant multivariate effects. No significant effects were detected for the physical exhaustion subscale of the EFI (see Table 2).

First, in regard to state anxiety, a significant main effect for Time was observed, $F(2, 98) = 21.06, p < .001$. Post hoc analysis revealed that significant reductions in state anxiety emerged 30 minutes following both aerobic exercise and quiet rest. However, the Condition $\times$ Time interaction approached significance, $F(2, 98) = 3.31, p = .04$. Examination of the ES in Table 2 revealed that reductions in state anxiety were associated with a substantially larger ES 30 minutes following exercise when compared to quiet rest. This finding indicates that, while both exercise and quiet rest were associated with reductions in state anxiety, larger effect sizes emerged during recovery from exercise.

Recent findings indicate that the use of the SAI in the context of acute exercise may be confounded by divergent response patterns between the anxiety and

Table 2 Effect Sizes (ES) for State Anxiety and the Exercise-Induced Feeling Inventory Subscales Before and After Exercise and Quiet Rest ($M \pm SD$)

Variable	Before	Post- 5 min	ES	Post- 30 min	ES
State Anxiety					
Exercise	33.7 (9.0)	31.0 (7.7)	-.32	27.8 (7.5)*	-.72
Quiet rest	33.0 (10.6)	31.1 (8.8)	-.20	29.9 (9.3)*	-.33
Positive Engagement					
Exercise	6.3 (2.8)	8.3 (2.9)**	.71	7.6 (3.2)**	.47
Quiet rest	6.7 (3.5)	6.2 (3.1)	-.15	6.3 (3.3)	-.12
Revitalization					
Exercise	3.9 (2.8)	7.9 (3.0)**	1.38	7.0 (3.2)**	1.03
Quiet rest	4.7 (3.4)	3.9 (2.9)	-.25	4.0 (3.0)	-.22
Tranquility					
Exercise	7.6 (2.7)	7.6 (3.2)	.00	8.9 (3.2)*	.43
Quiet rest	7.9 (3.3)	8.7 (2.8)	.26	8.9 (3.2)*	.32
Physical Exhaustion					
Exercise	4.2 (3.1)	3.2 (2.8)	-.34	4.0 (3.7)	-.06
Quiet rest	4.4 (3.3)	4.3 (3.2)	-.03	4.6 (3.4)	.05

Significantly different: *from baseline value, $p < .001$; **from baseline and quiet rest..

arousal laden items (Ekkekakis, Hall, & Petruzzello, 1999; Rejeski, Hardy, & Shaw, 1991). Therefore, changes in the individual SAI items were analyzed. Specifically, to allow for a direct comparison with the findings of Ekkekakis et al. (1999), the 10 items that comprise the short form of the SAI were analyzed using a repeated measures MANOVA. Results revealed a significant multivariate effect for Time (Wilks' lambda = .49), $F(20, 176) = 3.70, p < .001$. Follow-up univariate analyses with Bonferroni adjustments demonstrated that the calm, tense, nervous, relaxed, worried, at ease, and worrying-over-possible-misfortunes items were responsible for the significant multivariate effect. Tukey post hoc analyses revealed that significant reductions in the nervous, worried, and worrying-over-possible-misfortunes items were observed at both 5 and 30 minutes following exercise. Additionally, significant decreases in the calm, tense, relaxed, and at-ease items emerged 30 minutes following exercise. The descriptive statistics for the 10-item SAI are listed in Table 3, and mean changes in all 20 SAI items are summarized in Table 4.

Second, in regard to positive engagement, a significant Condition $\times$ Time interaction, $F(2, 98) = 24.44, p < .001$, was observed. Post hoc analyses revealed that positive engagement was significantly increased at both 5 and 30 minutes following only the exercise condition. Furthermore, positive engagement was found to be significantly higher following exercise when compared to quiet rest. Thus, improvements in positive engagement were only observed following exercise, and positive engagement was found to be significantly higher following exercise when compared to quiet rest.

Third, for revitalization, a significant Condition $\times$ Time interaction, $F(2, 98) = 44.19, p < .001$, was detected. Post hoc analyses revealed that significant increases in revitalization were observed 5 and 30 minutes following the exercise condition, and the increases were significantly larger than changes observed following quiet rest. Therefore, improvements in revitalization were only realized

Table 3 Selected Individual Items of SAI Scores Before and After Acute Exercise ($M \pm SD$)

Item	Before	Post-5	Post-30
1. Calm	1.94 (.74)	2.16 (.87)	1.46 (.65)*
2. Tense	1.68 (.84)	1.42 (.61)	1.24 (.43)*
3. Frightened	1.08 (.27)	1.02 (.14)	1.04 (.20)
4. Nervous	1.36 (.66)	1.08 (.27)*	1.08 (.34)*
5. Relaxed	2.22 (.76)	2.12 (.85)	1.62 (.81)*
6. Worried	1.62 (1.04)	1.26 (.53)*	1.28 (.61)*
7. At Ease	2.18 (.87)	1.92 (.85)	1.58 (.84)*
8. Jittery	1.26 (.49)	1.30 (.68)	1.08 (.27)
9. Steady	2.04 (.88)	1.94 (.82)	1.76 (.87)
10. Worrying	1.78 (1.05)	1.44 (.67)*	1.38 (.64)*

* Significantly lower than baseline value, $p < .005$.

Table 4 Changes in the State Anxiety Inventory Items Following Acute Exercise (*M* ± *SD*)

SAI Item	Post-5	Post-30
1. Calm	-0.22 (1.01)	0.48 (.81)
2. Secure	0.04 (.72)	0.20 (.64)
3. Tense	0.26 (1.06)	0.44 (.99)
4. Strained	-0.10 (1.03)	0.14 (.73)
5. At Ease	0.26 (.92)	0.60 (1.10)
6. Upset	0.22 (.65)	0.20 (.67)
7. Worrying	0.34 (.72)	0.40 (.78)
8. Satisfied	0.34 (.82)	0.36 (.72)
9. Frightened	0.06 (.24)	0.04 (.19)
10. Comfortable	0.04 (.86)	0.32 (.65)
11. Confident	0.16 (.62)	0.22 (.68)
12. Nervous	0.28 (.70)	0.28 (.75)
13. Indecisive	0.08 (.88)	0.26 (.60)
14. Jittery	-0.04 (.91)	0.18 (.59)
15. Relaxed	0.10 (.99)	0.60 (.86)
16. Content	0.32 (.71)	0.34 (.63)
17. Worried	0.36 (.85)	0.34 (.96)
18. Confused	0.12 (.32)	0.08 (.49)
19. Steady	0.10 (1.05)	0.28 (.88)
20. Pleasant	0.14 (.76)	0.18 (.63)

Note. Positive values = decrease in that item following exercise; negative values = increase in that item.

following exercise and the increases were found to be significantly larger than those observed following quiet rest.

Fourth, in regard to tranquility, a significant main effect for Time, $F(2, 98) = 8.83, p < .001$, was observed. The Condition $\times$ Time interaction approached significance, $F(2, 98) = 2.97, p = .05$. Post hoc analyses indicated that tranquility was significantly increased 30 minutes following exercise and quiet rest. Thus, both exercise and quiet rest were associated with improvements in tranquility.

Discussion

The principal finding of the present investigation indicated that acute bouts of aerobic exercise performed in a naturalistic setting, along with quiet rest, were associated with different affect responses. Specifically, whereas improvements in state anxiety and tranquility were observed following both exercise and quiet rest, increases in positive engagement and revitalization only emerged following exercise. Thus the findings provide only partial support for the *a priori* hypotheses.

Although the mechanisms (e.g., distraction, social environmental) that contributed to the differential feeling state responses cannot be determined at present,

the findings suggest that quiet rest may only be comparable to exercise for improving specific aspects of emotional health. Therefore, while both exercise and quiet rest help attenuate state anxiety and contribute to a sense of tranquility, exercise may be superior to more passive interventions in enhancing pleasant feeling states such as positive engagement and revitalization. Nevertheless, to substantiate the present assertions, more research is needed on the influence of different behavioral therapy conditions upon concomitant changes in positive and negative feeling states. Furthermore, studies examining the influence of bouts of naturalistic-setting exercise and quiet rest on other psychological states (e.g., affect and depression) are also needed to extend the present findings to other mental health outcomes.

In contrast to previous findings (Ekkekakis et al., 1999; Katula, Blissmer, & McAuley, 1999), divergence in the response patterns of the arousal and anxiety laden items of the SAI in the present study did not emerge 5 minutes following exercise. Nevertheless, differences between the studies in both exercise intensity and the timing of postexercise assessments may have contributed to these conflicting results. For example, whereas concomitant increases in arousal and decreases in anxiety items have been found immediately following bouts of high intensity exercise (Ekkekakis et al., 1999; Katula et al., 1999), the reduction in total SAI score observed by Katula et al. (1999) following a single episode of light intensity walking reflected decreases in both arousal and anxiety items. Thus, since the present study employed self-selected intensity which was subjectively rated as neither hard nor light, perhaps the exercise session was not demanding enough to elicit elevations in effort related tension that would persist for 5 minutes postexercise. It may be that the exercise-induced response divergence is influenced by the intensity of exercise and the proximity of the SAI administration to the cessation of activity (Ekkekakis & Petruzzello, 1999). Consequently, the impact of both the dose of activity and the timing of postexercise assessments on the interpretation of total SAI scores following exercise must be carefully considered in future exploration of the anxiolytic effect of acute exercise.

In light of the ambiguous findings evident in prior literature, the postexercise elevation in positive engagement is also of particular interest. It has been reported previously that there are increases in positive engagement following acute exercise in group environments (Gauvin & Rejeski, 1993; Turner et al., 1997), yet similar improvements have not been consistently found in laboratory environments (Gauvin & Rejeski, 1993; Rejeski et al., 1995). Thus it is reasonable to speculate that the social context inherent to the naturalistic exercise settings was integral to the elevation in positive engagement observed in the present study. Also, this finding emphasizes both the impact of the social environmental context of physical activity settings on psychological states and the value of conducting future research of the exercise/affect relationship in more naturalistic exercise environments.

The similarity of the responses to the positive engagement and revitalization subscales of the EFI in this study may lead some readers to question whether the subscales indeed measure distinct feeling states. Whereas this response pattern is consistent with those reported in some prior studies (Treasure & Newberry, 1998; Gauvin et al., 1996), other investigators have observed different responses in these subscales following acute exercise (Bozoian et al., 1994; Gauvin, Rejeski, & Reboussin, 2000; Rejeski et al., 1995). Moreover, when similar response patterns

have emerged, postexercise changes in revitalization have been consistently associated with larger effect sizes (Bozoian et al., 1994; Gauvin & Rejeski, 1993; Gauvin et al., 1996; Treasure & Newberry, 1998).

It is also important to consider that while some overlap in the subscales was detected in the initial validation of the EFI, revitalization and positive engagement also demonstrated different correlations with indices of positive affect, and each factor solution tested indicated that the subscales tap distinct feeling states (Gauvin & Rejeski, 1993). Obviously, the design of the current study does not permit us to determine whether the positive engagement and revitalization subscales of the EFI should be combined. Based on existing data, it appears that the behavior of these two subscales may depend on the research question and the social environmental context of the activity. Clearly, the evaluation of any inventory is an ongoing process and the psychometric properties of the EFI warrant further inquiry.

The present findings indicate that improvements in state anxiety and feeling states following exercise were unrelated to SPA. Therefore, the hypothesis that the psychological benefits of acute exercise would be moderated by SPA was not supported. Further inquiry is needed, however, to delineate the nature of the relationship between SPA and exercise-induced changes in psychological states. Katula et al. (1998) speculated that physique anxiety, while conceptualized as a stable psychological trait, may demonstrate labile properties within socially evaluative conditions. With this in mind, it may be that SPA serves as a salient determinant of psychological responses if the exercise environment provides sufficient threat to elicit self-presentational physique anxiety. Although the naturalistic environment in the present study provided the prospect for the realization of such negative physique evaluation, it cannot be determined that the high SPA women actually perceived the setting to be threatening. Therefore, to substantiate the contention that SPA does not moderate exercise-induced changes in psychological states, additional research is needed in which the evaluative threat of the environment is manipulated.

The differences in body composition among the high and low SPA participants also possess cogent health implications and warrant discussion. Consistent with previous research (Bane & McAuley, 1998), women with heightened SPA had higher % body fat and BMI than their less physique-anxious counterparts. Nevertheless, the mean body composition values for both the high and low SPA women were within age related norms. Because much prior SPA research has relied on indirect assessments of body composition, it is unclear whether the body composition values evident in the high SPA participants are typical of individuals with high physique anxiety. Hence, given the deleterious influence that self-presentational concerns about physique may perpetuate (Leary, Tchividjian, & Kraxberger, 1994), more research is warranted to discern the nature of the interaction between SPA, perceptions of body size and body composition, and health behaviors.

Given that the LTEQ data suggests the sample was composed of physically active women, the extent to which the findings can be generalized to sedentary adults is not yet clear. Provided that one's fitness level or prior physical activity participation may moderate the relationship between exercise and improvements in psychological well-being (Boutcher & Landers, 1988; Dishman, Farquhar, & Cureton, 1994), further exploration on psychological responses to exercise is needed in different population groups (i.e., obese, sedentary, or low fit individuals) in order to extend the present findings to more diverse populations.

In conclusion, while bouts of acute aerobic exercise and quiet rest are associated with comparable improvements in state anxiety and tranquility, aerobic exercise was superior in generating feelings of positive engagement and revitalization. Because improvements in state anxiety and positive feeling states can be viewed as beneficial to health, the present findings address pertinent public health issues. Specifically, current estimates suggest that the pandemic nature of psychological disorders have exceeded the capacity of the health care system (American Psychiatric Association, 1994), and this has stimulated interest in identifying other primary prevention strategies. The present findings suggest that exercise could function as one such alternative (Morgan, 1997; Raglin, 1997); this finding is extended to women high in physique anxiety. Finally, the established role of physical activity in the prevention of chronic diseases (U.S. Dept. Health and Human Services, 1999), and the effect of aerobic exercise on positive feeling states as demonstrated in the present study, suggest that exercise is associated with unique health benefits. Therefore, the use of exercise as a form of primary prevention for psychological disturbance may be preferable to alternative options.

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